

# MongoDB Capstone Project 2: Hospital Appointment & Billing Analytics

## Assignment Tasks

### 1. Basic Queries

```
db.patients.insertMany([  
  { patient_id: 1, name: "Aarav", city: "Hyderabad", age: 29, gender: "Male" },  
  { patient_id: 2, name: "Priya", city: "Bangalore", age: 34, gender: "Female" },  
  { patient_id: 3, name: "Rahul", city: "Mumbai", age: 41, gender: "Male" },  
  { patient_id: 4, name: "Sneha", city: "Delhi", age: 26, gender: "Female" },  
  { patient_id: 5, name: "Kiran", city: "Hyderabad", age: 37, gender: "Male" },  
  { patient_id: 6, name: "Meera", city: "Chennai", age: 31, gender: "Female" }  
]);
```

```
db.doctors.insertMany([  
  { doctor_id: 101, name: "Dr. Sharma", specialization: "Cardiology",  
    consultation_fee: 1200, city: "Hyderabad" },  
  { doctor_id: 102, name: "Dr. Iyer", specialization: "Dermatology",  
    consultation_fee: 800, city: "Bangalore" },  
  { doctor_id: 103, name: "Dr. Khan", specialization: "Orthopedics",  
    consultation_fee: 1500, city: "Mumbai" },  
  { doctor_id: 104, name: "Dr. Reddy", specialization: "Pediatrics",  
    consultation_fee: 900, city: "Delhi" },  
  { doctor_id: 105, name: "Dr. Mehta", specialization: "Neurology",  
    consultation_fee: 2000, city: "Hyderabad" }  
]);
```

```
db.appointments.insertMany([
  { appointment_id: 1001, patient_id: 1, doctor_id: 101, visit_date: "2026-03-01", status: "Completed", bill_amount: 1200 },
  { appointment_id: 1002, patient_id: 2, doctor_id: 102, visit_date: "2026-03-02", status: "Completed", bill_amount: 800 },
  { appointment_id: 1003, patient_id: 1, doctor_id: 105, visit_date: "2026-03-04", status: "Pending", bill_amount: 2000 },
  { appointment_id: 1004, patient_id: 3, doctor_id: 103, visit_date: "2026-03-05", status: "Completed", bill_amount: 1500 },
  { appointment_id: 1005, patient_id: 5, doctor_id: 101, visit_date: "2026-03-07", status: "Cancelled", bill_amount: 1200 },
  { appointment_id: 1006, patient_id: 6, doctor_id: 104, visit_date: "2026-03-08", status: "Completed", bill_amount: 900 },
  { appointment_id: 1007, patient_id: 4, doctor_id: 104, visit_date: "2026-03-09", status: "Pending", bill_amount: 900 },
  { appointment_id: 1008, patient_id: 3, doctor_id: 105, visit_date: "2026-03-10", status: "Completed", bill_amount: 2000 }
]);
```

```
db.patients.find();
```

```
db.doctors.find();
```

```
db.appointments.find();
```

```
db.patients.find({city:"Hyderabad"});
```

```
db.doctors.find({specialization:"Cardiology"});
```

```
db.appointments.find({status:"Completed"});
```

## 2. Filtering Queries

```
db.patients.find({age:{$gt:30}});
```

```
db.doctors.find({consultation_fee:{$gt:1000}});
```

```
db.doctors.find({consultation_fee:{$gte:900,$lte:1600}});
```

```
db.appointments.find({bill_amount:{$gt:1000}});
```

```
db.appointments.find({status:{$ne:"Cancelled"}});
```

```
db.patients.find({city:{$in:["Hyderabad","Mumbai"]}});
```

```
db.doctors.find({city:{$in:["Hyderabad","Delhi"]}});
```

## 3. Projection Queries

```
db.patients.find({}, {name:1, city:1, _id:0});
```

```
db.doctors.find({}, {_id:0, name:1, specialization:1, consultation_fee:1});
```

```
db.appointments.find({}, {_id:0, appointment_id:1, status:1, bill_amount:1});
```

## 4. Sorting, Limit, Skip

```
db.doctors.find().sort({consultation_fee:1});
```

```
db.doctors.find().sort({consultation_fee:-1});
```

```
db.doctors.find().sort({consultation_fee:-1}).limit(3);
```

```
db.doctors.find().sort({consultation_fee:1}).limit(2);
```

```
db.patients.find().skip(2);
```

```
db.patients.find().sort({age:-1});
```

## 5. Update Operations

```
db.doctors.updateOne({name:"Dr. Sharma"},{$set:{consultation_fee:1300}});
```

```
db.appointments.updateMany({status:"Pending"},{$set:{priority:"High"}});
```

```
db.doctors.updateMany({city:"Hyderabad"},{$set:{available:"true"}});
```

```
db.patients.updateOne({name:"Meera"},{$set:{city:"Bangalore"}});
```

## 6. Delete Operations

```
db.doctors.deleteOne({name:"Dr. Iyer"});
```

```
db.appointments.deleteMany({status:"Cancelled"});
```

```
db.patients.deleteMany({city:"Delhi"});
```

## 7. Count Queries

```
db.patients.countDocuments();
```

```
db.doctors.countDocuments({city:"Hyderabad"});
```

```
db.appointments.countDocuments({status:"Completed"});
```

```
db.patients.countDocuments({city:"Hyderabad"});
```

## 8. Aggregation Queries

```
db.doctors.aggregate([{\n  $group:{\n    _id:"$specialization",\n    avg_fee:{$avg:"$consultation_fee"}\n  }\n}]);
```

```
db.doctors.aggregate([{\n  $group:{\n    _id:null,\n    max_fee:{$max:"$consultation_fee"}\n  }\n}]);
```

```
}  
});
```

```
db.doctors.aggregate([  
  $group:{  
    _id:null,  
    min_fee:{$min:"$consultation_fee"}  
  }  
]);
```

```
db.appointments.aggregate([  
  $group:{  
    _id:"$status",  
    total_amount:{$sum:"$bill_amount"}  
  }  
]);
```

```
db.appointments.aggregate([  
  $group:{  
    _id:"$doctor_id",  
    total_appointment:{$sum:1}  
  }  
]);
```

```
db.appointments.aggregate([  
  $group:{  
    _id:"$patient_id",  
    total_appointment:{$sum:1}  
  }  
]);
```

```
db.patients.aggregate([  
  $group:{  
    _id:"$city",  
    avg_age:{$avg:"$age"}  
  }  
]);
```

```
db.appointments.aggregate([  
  $group:{  
    _id:"$doctor_id",  
    total_amount:{$sum:"$bill_amount"}  
  }  
]);
```

```
db.appointments.aggregate([  
  $lookup:{
```

```
    from:"patients",
    localField:"patient_id",
    foreignField:"patient_id",
    as:"patient_details"
  }
},
{$unwind:"$patient_details"},
{
  $group:{
    _id:"$patient_details.city",
    total_amount:{$sum:"$bill_amount"}
  }
}
]);
```

## 9. Multi-Collection Queries using \$lookup

```
db.appointments.aggregate([{$lookup:{
  from:"patients",
  localField:"patient_id",
  foreignField:"patient_id",
  as:"patient_details"
}
}]);
```



```
db.appointments.aggregate([  
  $lookup:{  
    from:"doctors",  
    localField:"doctor_id",  
    foreignField:"doctor_id",  
    as:"doctor_details"  
  }  
]);
```

```
db.appointments.aggregate([  
  $lookup:{  
    from:"patients",  
    localField:"patient_id",  
    foreignField:"patient_id",  
    as:"patient_details"  
  }  
],  
{ $unwind:"$patient_details"},
```

```
{  
  $lookup:{  
    from:"doctors",  
    localField:"doctor_id",  
    foreignField:"doctor_id",
```

```
    as:"doctor_details"
  }
},
{$unwind:"$doctor_details"},
{
  $project:{
    _id:0,
    patient_name:"$patient_details.name",
    doctor_name:"$doctor_details.name",
    appointment_id:1
  }
}

]);
```

```
db.appointments.aggregate([
  $lookup:{
    from:"patients",
    localField:"patient_id",
    foreignField:"patient_id",
    as:"patient_details"
  }
},
{$unwind:"$patient_details"},
```

```
{
  $lookup:{
    from:"doctors",
    localField:"doctor_id",
    foreignField:"doctor_id",
    as:"doctor_details"
  }
},
{$unwind:"$doctor_details"},
{
  $project:{
    _id:0,
    patient_name:"$patient_details.name",
    city:"$patient_details.city",
    doctor_name:"$doctor_details.name",
    specialization:"$doctor_details.specialization",
    status:1,
    bill_amount:1
  }
}

});
```

```
db.patients.aggregate([{\n  $lookup:{\n    from:"appointments",\n    localField:"patient_id",\n    foreignField:"patient_id",\n    as:"appointments"\n  }\n}]);
```

```
db.doctors.aggregate([{\n  $lookup:{\n    from:"appointments",\n    localField:"doctor_id",\n    foreignField:"doctor_id",\n    as:"appointments"\n  }\n}]);
```

## 10. Advanced Aggregation

```
db.appointments.aggregate([{\n  $group:{\n    _id:"$doctor_id",\n    total_revenue:{$sum:"$bill_amount"}\n  }\n}]);
```

```
db.appointments.aggregate([  
  $lookup:{  
    from:"doctors",  
    localField:"doctor_id",  
    foreignField:"doctor_id",  
    as:"doctor_details"  
  }  
},  
{$unwind:"$doctor_details"},  
{  
  $group:{  
    _id:"$doctor_details.specialization",  
    total_revenue:{$sum:"$bill_amount"}  
  }  
}  
]);
```

```
db.appointments.aggregate([  
  { $group: { _id: "$patient_id", total: { $sum: "$bill_amount" } } },  
  { $sort: { total: -1 } },  
  { $limit: 1 }
```

```
]);
```

```
db.appointments.aggregate([  
  { $group: { _id: "$doctor_id", count: { $sum: 1 } } },  
  { $sort: { count: -1 } },  
  { $limit: 1 }  
]);
```

```
db.appointments.aggregate([  
  {  
    $match:{status:"Completed"}  
  },  
  {  
    $group:{_id:null, total_revenue:{ $sum:"$bill_amount"}}  
  }  
]);
```

```
db.appointments.aggregate([  
  { $lookup: { from: "patients", localField: "patient_id", foreignField:  
"patient_id", as: "p" } },  
  { $unwind: "$p" },  
  { $group: { _id: "$p.city", count: { $sum: 1 } } },  
  { $sort: { count: -1 } },  
  { $limit: 1 }  
]);
```

```
db.appointments.aggregate([
  { $lookup: { from: "patients", localField: "patient_id", foreignField:
"patient_id", as: "p" }},
  { $unwind: "$p" },
  { $group: { _id: "$p.city", total: { $sum: "$bill_amount" } } },
  { $sort: { total: -1 } },
  { $limit: 1 }
]);
```

```
db.appointments.aggregate([
  { $lookup: { from: "doctors", localField: "doctor_id", foreignField: "doctor_id",
as: "d" }},
  { $unwind: "$d" },
  { $group: { _id: "$d.specialization", avg_bill: { $avg: "$bill_amount" } } },
  { $sort: { avg_bill: -1 } },
  { $limit: 1 }
]);
```